[INCH-POUND]
A-A-59393
17 April 2000
SUPERSEDING
MIL-M-80184B
18 December 1990
MIL-M-80212B
26 June 1986

COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, VERTICAL MULTISPINDLE, DOUBLE COLUMN, COMPUTER NUMERICAL CONTROL (CNC)

The General Services Administration has authorized the use of this commercial item for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC), double column, vertical multispindle, milling machines with fixed columns and traveling beds intended for use in general and specialty machine shops, large industrial facilities, and high-volume manufacturing plants. The milling machines are intended for multiaxis contour milling, boring, drilling, reaming, and tapping cast iron, steel, titanium, aluminum, glass tiles, honeycomb laminates, graphite epoxies, and advanced composite materials.

2. CLASSIFICATION. The milling machines shall be of the following types, classes, styles, and sizes. The type, class, style, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Single spindle
Type II - Two spindles
Type III - Three spindles
Type IV - Four spindles

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3417

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

Class 1 - 20 to 25 horsepower
Class 2 - 30 to 40 horsepower
Class 3 - 50 to 60 horsepower
Class 4 - 75 to 100 horsepower
Class 5 - 125 horsepower and greater

Style A - 3 axes, full contouring
Style B - 4 axes, full contouring
Style C - 5 axes, full contouring

Size - Sizes are shown in table I

TABLE I. Sizes of milling machines.

Size	Class	Minimum machine travel, inches		
		X-axis	Y-axis	Z-axis
1	1	138	152	40
2		157	171	40
3		217	197	39
4		256	197	39
5		551	149	53
6		648	222	59
7		648	261	79
8	2	120	60	28
9		120	78	28
10		144	60	28
11		144	78	28
12		168	60	28
13		168	78	28
14	3	128	66	18
15		530	202	39
16	4	228	64	28
17		228	114	28
18		228	124	28
19		228	164	28
20		252	408	168
21	5	228	104	28
22		228	154	28
23		530	231	59

3. SALIENT CHARACTERISTICS

3.1 General requirements. The milling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The machine shall be constructed from materials with sufficient mass and strength and configured in a shape and size that shall fully support the workpiece and all required machine components while

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

minimizing element deflections and initiated and/or transmitted vibration. All parts subject to wear, breakage, or distortion shall be easily accessible for adjustment, replacement, or repair. The milling machine and CNC system shall be compatible and integrated in design as a functional unit. The CNC system shall accept workpiece machining program instructions and shall direct all tool changes, spindle accessory changes, spindle head operation, and linear-rotational motion on all axes for continuous-path, full-contouring machining. The term "full contouring" is defined herein as a closed-loop control system having simultaneous control of position, direction, and speed of traverse by linear, parabolic, circular, and helical interpolation. The CNC system shall be capable of machine operator manual control by hand controls and switches, semiautomatic control by manual data input (MDI), and automatic control by programmable machining operation. The CNC system shall always be active when the milling machine is powered up.

3.1.1 Identification of axes. Unless otherwise specified (see 7.2(c)), axes of the milling machine shall be identified in accordance with Electronics Industries Association (EIA) RS-267.

3.1.2 Rapid traverse speed. The X-, Y-, and Z-axis rapid traverse speeds shall be as specified (see 7.2(d)).

3.2 Components. The milling machine shall include, as a minimum, the following components:

3.2.1 Base. The milling machine shall have a base that supports the worktable. The base shall have a means for leveling and anchoring the machine to the floor or foundation.

3.2.2 Worktable. Unless otherwise specified (see 7.2(e)), the length, width, and workpiece weight capacity of the worktable shall be that considered by the manufacturer as standard. Unless otherwise specified (see 7.2(f)), the worktable surface shall be provided with a workpiece hold-down system that is considered by the manufacturer as standard. If required, the worktable shall have the number and size of T-slots as specified (see 7.2(g)). When provided, T-slot dimensions shall conform to the requirements of American National Standards Institute (ANSI) B5.1. When specified (see 7.2(h)), an optional worktable surface wear plate shall be furnished.

3.2.3 Columns. The milling machine shall have two fixed vertical columns. The columns shall be located on opposite sides of the worktable and shall be attached either to the base or mounted on the floor or foundation.

3.2.4 Crossrail. The milling machine shall have a crossrail mounted to the columns that spans the worktable and provides support for the spindles. If required, the crossrail shall have the minimum vertical travel (W-axis) as specified (see 7.2 (i)).

3.2.5 Spindle assembly. The spindle assembly shall be that considered by the manufacturer as standard. Unless otherwise specified (see 7.2(j)), the spindle(s) shall be provided with a means of cooling to control spindle thermal growth. Unless otherwise specified (see 7.2(k), spindles shall be independently programmable for movement and operation. If required, the following characteristics of each spindle assembly provided shall be as specified (see 7.2(l)):

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

a. Mounting location.
b. Drive motor horsepower.
c. Speed range.
d. Spindle nose configuration.
e. Tool retention mechanism.
f. Tool taper size.
g. Tool shank V-flange.

3.2.6 Rigid tapping. Unless otherwise specified (see 7.2(m)), rigid tapping shall be provided.

3.2.7 Machineways. The milling machine shall have precision machineways that are hardened to resist wear and are match ground and scraped for accuracy. Machineways shall be replaceable and bolted into position to provide for ease of maintenance, adjustment, or replacement. All machineways shall have telescoping protective covers and/or wipers to keep chips, coolant, and other debris from the bearing surfaces.

3.2.8 Automatic tool changer and storage magazine. If required, the milling machine shall have an automatic tool changer and tool storage magazine as specified (see 7.2(n)). The automatic tool changer shall be capable of selecting the next tool from the storage magazine using the bidirectional random shortest path method. The automatic tool changer shall be capable of exchanging tools, returning the previous tool to the storage magazine, and tracking the tool for reuse. A means shall be provided to shield the stored tools from dirt and other debris during machine operation.

3.2.9 Automatic accessory head changer and storage magazine. If required, the milling machine shall have one or more spindle accessory heads, an automatic accessory head changer, and an accessory head storage magazine as specified (see 7.2(o)). A means shall be provided to shield the stored accessory heads from dirt and other debris during machine operation. Regardless of the number of accessory heads specified, a protective spindle ram cover plate shall be provided if the machine spindle ram is designed for accessory heads.

3.2.10 Pressurized air system. The milling machine shall be provided with an air supply system that is connected to either shop air or a dedicated air compressor and tank system as specified (see 7.2(p)). Piping, tubing, fittings, filters, dryers, and other appurtenances necessary for the required operation of the compressed air system shall be included by the manufacturer. The piping system shall direct pressurized air through the spindle and spindle head(s), if provided, and to tool adapters to remove debris from the taper surfaces during the spindle head- and tool-changing processes.

3.2.11 Chip conveyor system. The milling machine shall have a means to automatically remove workpiece chips from the work zone. Unless otherwise specified (see 7.2(q)), a conveyor system shall be provided to transfer chips from the machine to a collection bin. The conveyor system shall include all drives, belts, idlers, and other necessary components required to function as a complete system and interface with the milling machine. The chip conveyor chute discharge height shall be as specified (see 7.2(r)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.12 Safety devices. An automatic stop circuit shall be provided to stop the feed motion and operation of spindle drives if a spindle motor becomes overloaded, overheated, or exceeds other operating parameters. When specified (see 7.2(s)), torque limiting devices shall be included on all drive systems. Limit switches shall be provided at both ends of each axis travel to prevent overtravel. The worktable, crossrail, cross saddle, and vertical saddle drives shall automatically switch to manual servocontrol during all safety interruptions. Automatic machine-actuated safety warnings, trips, and/or stoppages will activate an indicator light(s) on the operator control panel and produce an audible alarm. Unless otherwise specified (see 7.2(t)), the milling machine shall be provided with a protective enclosure to isolate the work zone. The enclosure shall be configured to permit operator access to and observation of the work zone.

3.2.13 CNC system. The CNC system shall be based on either a personal computer (PC) open-architecture or other software-programmable microprocessor design. System hardware and control panels shall be industrial grade. Unless otherwise specified (see 7.2(u)), control panels shall be industrial grade type 12 enclosures conforming to National Electrical Manufacturers Association (NEMA) 250.

3.2.13.1 Hardware. Unless otherwise specified (see 7.2(v)), the CNC system shall include, as a minimum, the following hardware:

a. One 32-bit, 16-megahertz microprocessor, math coprocessor, and 32-bit local buss.
b. Buffer capacity of 128 kilobytes.
c. RAM storage capacity of 32 megabytes.
d. A 250-megabyte hard disk drive.
e. A 3.5-inch, 1.4-megabyte floppy disk drive.
f. An IEEE 1284 parallel port.
g. Two EIA RS-232 serial ports.
h. A 14-inch color monitor.
i. A manual data input (MDI) device consisting of either a 101-key computer keyboard or an alpha-numeric keypad with dedicated keys for specific machine functions.

3.2.13.2 Control panel. Manual controls shall be provided that are hardwired to override or bypass CNC machine instructions. Unless otherwise specified (see 7.2(w)), the CNC system shall include, as a minimum, a control panel with the following manual controls and switches:

a. Emergency stop button.
b. Key-actuated system lock.
c. Power on/off button.
d. Spindle start/stop buttons.
e. Spindle speed override rotary switches.
f. Feed hold button.
g. Feed rate override rotary switches.
h. Jog and jog direction joy stick.
i. CNC system start/stop buttons (cycle start/stop switches).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.13.3 Machine status indicators. Unless otherwise specified (see 7.2(x)), the CNC system control panel shall include, as a minimum, the following machine status indicators:

a. Power on indicator.
b. Operational mode status indicator.
c. Axis movement indicator.
d. Spindle power indicators.
e. Spindle speed indicators.
f. Spindle overload indicator.
g. Overload and/or trouble indicator.
h. Overload and/or trouble audible alarm.
i. Accessory system status indicator(s) (e.g., air, coolant, lubrication).

3.2.13.4 Pendant control. Unless otherwise specified (see 7.2(y)), a swing arm-supported control or hand-held pendant control shall be provided in addition to the main CNC system control panel. The pendant control shall allow the operator to override the CNC system and manually control basic machine functions from the work zone during workpiece setup and tool path checks.

3.2.13.5 Operating modes. Unless otherwise specified (see 7.2(z)), the CNC system shall include, as a minimum, the following operating modes:

a. Manual operation mode.
b. Manual data input (MDI) mode.
c. Setup mode.
d. Transmit and receive mode.
e. Programming mode.
f. Conversational mode.
g. Automatic mode.
h. Split screen program preview mode.
i. Background editing mode.
j. Machine diagnostic/maintenance/monitoring modes.

3.2.13.6 Axis and tool management. Unless otherwise specified (see 7.2(aa)), the CNC system shall include, as a minimum, the following axis and tool management features:

a. Tool length compensation.
b. Tool life management.
c. Cutter diameter compensation.
d. Tool offsets.
e. Number of tool offsets.
f. Tool breakage detection.
g. Scale feedback.
h. Drive ball screw backlash compensation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.2.13.7 Software features. Unless otherwise specified (see 7.2(bb)), the CNC system shall include, as a minimum, the following software programming and control features:

a. EIA RS-274 variable data block programming format (standard G-, M-, T-codes).
b. EIA RS-494 programming format (32 bit binary and 7 bit ASCII).
c. Decimal point programming.
d. Inch/metric programming.
e. Absolute and incremental motion control programming.
f. Cartesian/polar coordinate programming.
g. Parametric programming.
h. Programmable zero-point reference.
i. Simultaneous control of linear and rotary axes.
j. Linear, circular, parabolic, and helical tool path interpolation.
k. Rigid tapping capability.
l. Manufacture's standard suite of preprogrammed machine operational cycles.
m. Three-dimensional imaging of the workpiece.
n. Workpiece graphic display rotation.
o. Tool path graphic display.
p. Maintenance monitoring package.
q. Distributive numerical control.
r. Diagnostics assistance.

3.2.14 Battery backup. When specified (see 7.2(cc)), the CNC system shall have a volatile memory battery backup or uninterrupted power supply (UPS) with a storage capacity of not less than 72 hours.

3.2.15 Lubrication system. Unless otherwise specified (see 7.2(dd)), the milling machine shall be supplied with an automatic lubrication system to direct a flow of lubricant from a reservoir to all machine components requiring continuous lubrication. The lubrication system shall include a reservoir, all necessary piping, pumps, motors, filters, strainers, heat exchangers, and fluid level indicators. A high-temperature alarm and a low-level alarm shall be provided. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.16 Coolant system. Unless otherwise specified (see 7.2(ee)), the milling machine shall be supplied with an automatic cutting tool coolant system. The coolant system, when actuated, shall direct a flow of coolant onto and/or around the area of the cutting tool at a selectable controlled rate of flow. The cooling system shall include a reservoir, all necessary piping, pumps, motors, filters, strainers, heat exchangers, and fluid level indicators. A high-temperature alarm and a low-level alarm shall be provided.

3.2.17 Electrical system. Unless otherwise specified (see 7.2(ff)), the electrical system shall conform to the ANSI/National Fire Protection Association (NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

(see 7.2(gg)). Unless otherwise specified (see 7.2(hh)), the CNC system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.18 Electric motors. The spindle drive motor horsepower shall correspond to a 30-minute duty cycle rating. The motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the NEMA MG-1. Unless otherwise specified (see 7.2(ii)), fractional horsepower motors shall have totally enclosed nonventilated (TENV) enclosures and integral horsepower motors shall have totally enclosed fan-cooled (TEFC) enclosures. Motion and position control motors and controls shall conform to NEMA ICS 1, NEMA ICS 3, and NEMA MG-7.

3.2.19 Optional accessories. Optional accessories shall be provided as specified (see 7.2(jj)).

3.3 Performance. Unless otherwise specified (see 7.2(kk)), the milling machine shall meet the performance requirements shown in table II when tested in accordance with section 5.3.3.

TABLE II. Performance requirements.

Test	Parameter	Performance
Linear displacement accuracy[1]	X, Y, Z, and W directions	0.0015 inch per 40 inches and 0.003 inch overall
Bidirectional repeatability	X, Y, Z, and W directions	±0.0003 inch
Volumetric performance[1]	Diagonal displacement	0.003 inch per 40 inches and 0.006 inch overall
Contouring performance (up to 80 percent maximum speed)	X-Y plane, 360 degrees X-Z, X-W planes, 190 degrees Y-Z, Y-W planes, 190 degrees	0.0015 inch
Squareness error (CW, CCW)	X-Y, X-Z, Y-Z, X-W, Y-W planes	5 arcseconds
Scale mismatch (CW, CCW) (ball bar length equal to 1/2 nominal work zone)	X- and Y-axes	0.001 inch
Periodic angular positioning accuracy	A-, B-, and C-axes	10 arcseconds
Bidirectional angular positioning repeatability	A-, B-, and C-axes	5 arcseconds
Spindle dynamics	Radial error – average	0.0003 inch
	Radial error – asynchronous	0.0005 inch
Spindle thermal growth (long term, more than 4 hours)	Without spindle chiller	0.006 inch
	With spindle chiller	0.003 inch

[1]If the axis travel is less than 40 inches, the tolerance shall be decreased proportionately.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(ll)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.8 and the OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(mm)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(nn)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(oo)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(pp)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2. Inspection. The milling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(qq)), the milling machine shall be tested in accordance with sections 5.3.1 through 5.3.7. Should a malfunction or failure occur during any test, the problem shall be corrected, and the test shall be rerun from the beginning.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(rr)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(ss)). Unless otherwise specified (see 7.2(tt)), the milling machine shall be tested in accordance with environmental conditions specified in American Society of Mechanical Engineers (ASME) B5.54.

5.3.2 Operational test. The milling machine and CNC system shall be operated in accordance with the manufacturer's standard operating procedures. The milling machine shall be operated for not less than 8 hours during which proper operation of all controls and features shall be verified.

5.3.3. Performance tests. Performance tests shall be conducted in accordance with the procedures specified in section 5 of ASME B5.54. The results of all tests shall meet the performance requirements shown in table II.

5.3.4. Spindle idle run loss test. A spindle idle run power loss test shall be performed on each spindle of the milling machine in accordance with section 7 of ASME B5.54. The measured power loss of the spindle(s) shall not exceed 20 percent of the spindle nominal rated power.

5.3.5 Chatter limit test. A chatter limit test shall be performed on each spindle of the milling machine in accordance with section 7 of ASME B5.54. Each spindle shall operate chatter free at all cutting loads throughout the total range of available torque, speed, and power in all axes of movement.

5.3.6 Deflection tests. Deflection tests shall be performed on each spindle of the milling machine in accordance with section 7 of ASME B5.54. The maximum deflection determined in either the end milling or the face milling tests shall not exceed the maximum deflection allowance shown in table III for the specific operation and test cut orientation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

Table III. Maximum deflection allowance.

Operation	Orientation of test cut	Axis	Maximum deflection, inch
End milling	Up	X	0.0003
	Down		0.0006
	Up	Y	0.0006
	Down		0.0008
	Up	Z	0.0002
	Down		0.0002
Face milling	Slit	X	0.0003
	1/2 Up		0.0002
	1/2 Down		0.0003
	Slit	Y	0.0003
	1/2 Up		0.0002
	1/2 Down		0.0003

5.3.7 Optional tests. Optional tests shall be conducted as specified (see 7.2(uu)).

5.4 Acceptance. Unless otherwise specified (see 7.2(vv)), preliminary acceptance tests shall be conducted at the manufacturer's site and final tests shall be conducted at the installation site. All tests shall be completed without making any adjustments to or performing any maintenance on the machine. Failure of the machine to meet any of the listed tests and the performance requirements shown in table II shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (see7.2(ww)).

7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B5.1 T-Slots, Their Bolts, Nuts, and Tongues

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

ANSI B11.8	Safety Requirements for Construction, Care, and Use of Drilling, Milling, and Boring Machines and Machine Tools

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

American Society of Mechanical Engineers (ASME)

ASME B5.54	Methods for Performance Evaluation of Computer Numerically Controlled Machining Centers

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

Electronic Industries Association (EIA)

RS-232	Interface Between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Exchange
RS-267	Axis and Motion Nomenclature for Numerically Controlled Machines
RS-274	Interchangeable Variable Block Data Format for Positioning and Contouring/Positioning Numerically Controlled Machines
RS-494	32 bit Binary CL (BCL) and 7 Bit ASCII CL (ACL) Exchange Input Format for Numerically Controlled Machines

Applications for copies should be sent to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

Institute of Electrical and Electronic Engineers (IEEE)

IEEE 1284	Standard Signaling Method for a Bidirectional Parallel Peripheral Interface for Personal Computers

Applications for copies should be sent to the Institute of Electrical and Electronics Engineers Inc., 345 East 47th Street, New York, NY 10017.

National Electrical Manufacturer's Association (NEMA)

NEMA 250	Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ICS 1	Industrial Control and Systems General Requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

NEMA ICS 3 Industrial Systems

NEMA MG-1 Motors and Generators

NEMA MG-7 Motion/Position Control Motors and Controls

Applications for copies should be sent to the National Electrical Manufacturer's Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

U.S. Department of Labor, Occupational Safety and Health Administration (OSHA)

OSHA 1910.212 Occupational Safety and Health Standards for General Industry, (29 CFR 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Ave. NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, class, style, and size of milling machine required (see 2).
c. Identification of axis, if different (see 3.1.1).
d. Rapid traverse speed, as specified (see 3.1.2).
e. Worktable length, width, and weight capacity, if different (see 3.2.2).
f. Workpiece hold-down system, if different (see 3.2.2).
g. Number and size of T-slots, as specified (see 3.2.2).
h. Worktable wear plate, if required (see 3.2.2).
i. Vertical travel of crossrail, as specified (3.2.4).
j. Spindle cooling, if different (see 3.2.5).
k. Spindle motion and operation independently programmable, if different (see 3.2.5).
l. Spindle characteristics, as specified (see 3.2.5).
m. Rigid tapping, if different (see 3.2.6).
n. Automatic tool changer and storage magazine, as specified (see 3.2.8).
o. Automatic accessory head changer and storage magazine, as specified (see 3.2.9).
p. Pressurized air system, as specified (see 3.2.10).
q. Chip conveyor system, if different (see 3.2.11).
r. Chip conveyor chute discharge height, as specified (see 3.2.11).

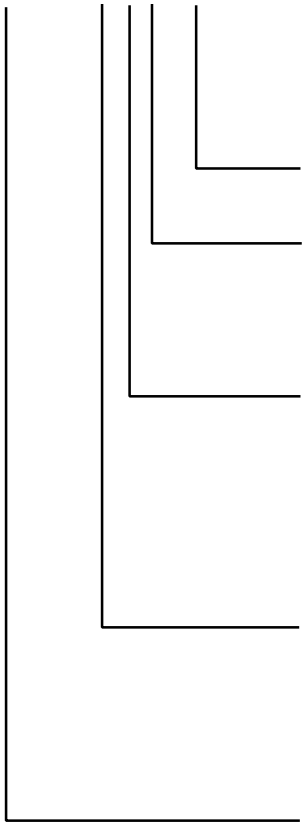
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

s. Drive system torque limiting devices, if required (see 3.2.12).
t. Protective enclosures, if different (see 3.2.12).
u. CNC system control panel enclosure, if different (see 3.2.13).
v. CNC system hardware, if different (see 3.2.13.1).
w. CNC system control panel manual controls and switches, if different (see 3.2.13.2).
x. CNC system machine status indicators, if different (see 3.2.13.3).
y. CNC system pendant control, if different (see 3.2.13.4).
z. CNC system operating modes, if different (see 3.2.13.5).
aa. CNC axis and tool management, if different (see 3.2.13.6).
bb. CNC system software features, if different (see 3.2.13.7).
cc. Battery backup, if required (see 3.2.14).
dd. Lubrication system, if different (see 3.2.15).
ee. Coolant system, if different (see 3.2.16).
ff. Electrical system industry standard, if different (see 3.2.17).
gg. Electrical system voltage, as specified (see 3.2.17).
hh. CNC system voltage, if different (see 3.2.17).
ii. Electric motors, if different (see 3.2.18)
jj. Optional accessories, as specified (see 3.2.19).
kk. Performance, if different (see 3.3).
ll. Machine dimensions and weight, as specified (see 3.4).
mm. System of units required (see 3.6).
nn. Dual system of units, if required (see 3.6).
oo. Nameplate, if different (see 3.7).
pp. Lubrication plate or chart, if required (see 3.8).
qq. Testing, if different (see 5.3).
rr. Test responsibility, if different (see 5.3.1).
ss. Certification of calibration, as specified (see 5.3.1).
tt. Test environmental conditions, if different (see 5.3.1).
uu. Optional tests, as specified (see 5.3.7).
vv. Acceptance, if different (see 5.4).
ww. Packaging requirements, as specified (see 6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

AA59393-XXXXX



Size: Sizes are shown in table I

Style:
A - Style A - 3 axes, full contouring
B - Style B - 4 axes, full contouring
C - Style C - 5 axes, full contouring

Class:
1 - Class 1 - 20 to 25 horsepower
2 - Class.2 - 30 to 40 horsepower
3 - Class 3 - 50 to 60 horsepower
4 - Class 4 - 75 to 100 horsepower
5 - Class 5 - 125 horsepower and greater

Type:
A - Type I - Single spindle
B - Type II - Two spindles
C - Type III - Three spindles
D - Type IV - Four spindles

CID number

MILITARY INTERESTS:

Custodians
Air Force - 99
Army - AL
Navy - SH

Reviewers
Air Force - 84
Navy - MC

PREPARING ACTIVITY
DLA - GS

(Project No. 3417-0214)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.